



量子计算和量子信息

1. 基本概念

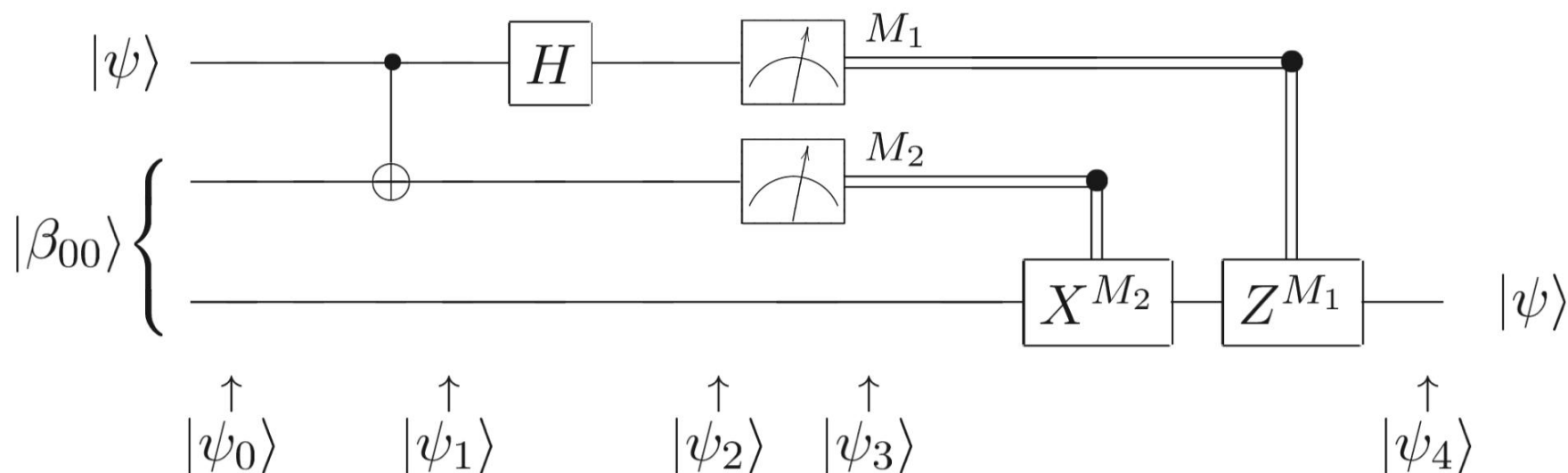
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海南大学理论物理研究中心

2024年春季学期

1.9 示例：量子隐形传态

结合前面学到的基本单元，考虑一个有实际应用价值的量子电路：
Quantum Teleportation。



把一个量子态转移到另外的地方去。
这个过程叫“Teleportation”。

1.9 示例：量子隐形传态

实验解读

Experimental quantum teleportation

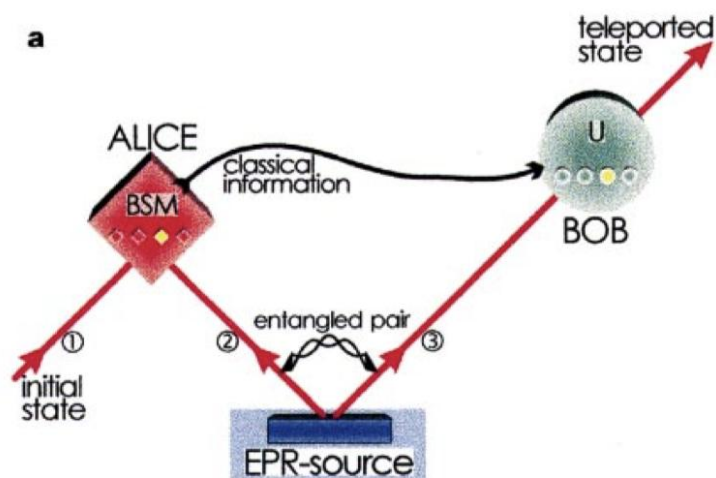
Dik Bouwmeester, Jian-Wei Pan, Klaus Mattle, Manfred Eibl, Harald Weinfurter & Anton Zeilinger

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Quantum teleportation—the transmission and reconstruction over arbitrary distances of the state of a quantum system—is demonstrated experimentally. During teleportation, an initial photon which carries the polarization that is to be transferred and one of a pair of entangled photons are subjected to a measurement such that the second photon of the entangled pair acquires the polarization of the initial photon. This latter photon can be arbitrarily far away from the initial one. Quantum teleportation will be a critical ingredient for quantum computation networks.

文章摘要用一句话概括了用光子做隐形传态到底是怎么实现的。

1.9 示例：量子隐形传态



以光子偏振来编码0, 1

$$|\psi\rangle = \alpha|\leftrightarrow\rangle + \beta|\uparrow\rangle$$

(1)

自发参量下转换，
生成光子2、3纠缠态

beta_11

$$|\psi^-\rangle_{23} = \frac{1}{\sqrt{2}}(|\leftrightarrow\rangle_2|\uparrow\rangle_3 - |\uparrow\rangle_2|\leftrightarrow\rangle_3)$$

(2)

对第1、2个光子在一个Bell基下测量：
(1/4的概率成功)

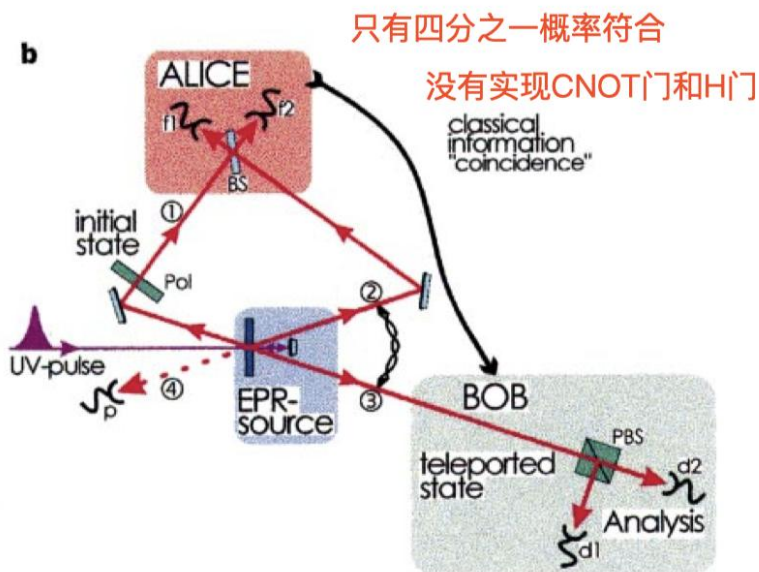
$$|\psi^-\rangle_{12} = \frac{1}{\sqrt{2}}(|\leftrightarrow\rangle_1|\uparrow\rangle_2 - |\uparrow\rangle_1|\leftrightarrow\rangle_2)$$

(3)

如果测量成功，则第三个光子态变为
(第一个光子的初始态)：

$$|\psi\rangle_3 = \alpha|\leftrightarrow\rangle_3 + \beta|\uparrow\rangle_3$$

(4)



1.9 示例：量子隐形传态

自发参量下转换，
生成光子2、3纠缠态

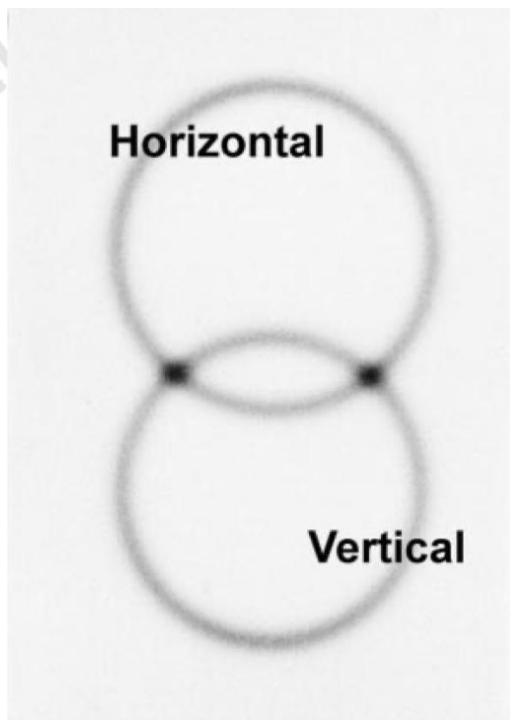
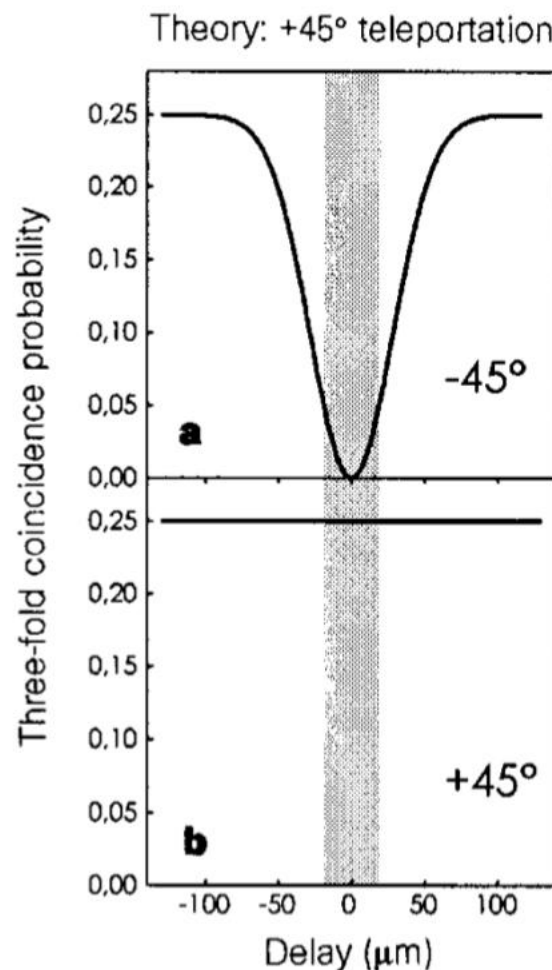


Figure 2 Photons emerging from type II down-conversion (see text). Photograph taken perpendicular to the propagation direction. Photons are produced in pairs. A photon on the top circle is horizontally polarized while its exactly opposite partner in the bottom circle is vertically polarized. At the intersection points their polarizations are undefined; all that is known is that they have to be different, which results in entanglement.

初始第一个光子45度偏振，
探测第三个光子的偏振。



1.9 示例：量子隐形传态

实验解读 （2017, Nature。墨子号总工程师 王建宇）

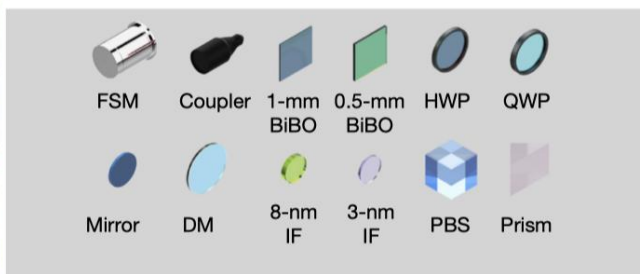
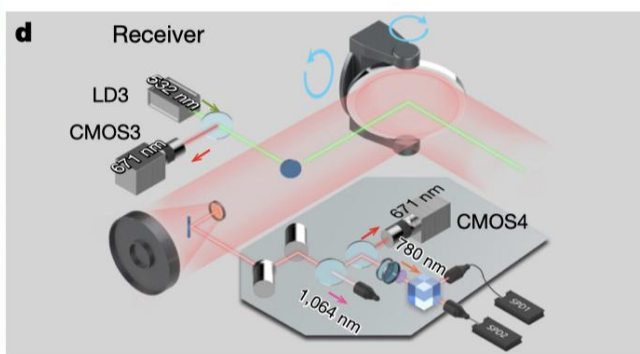
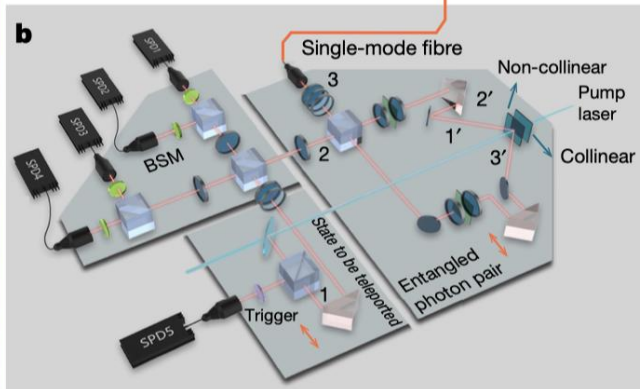
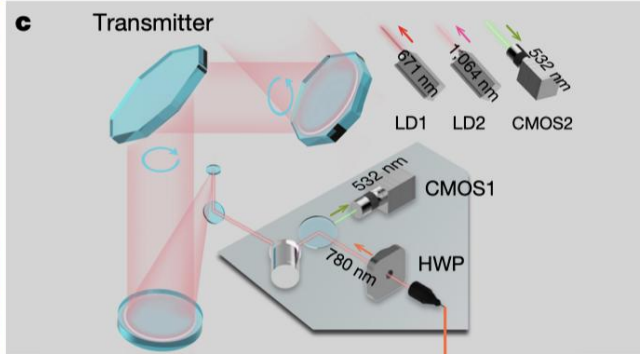
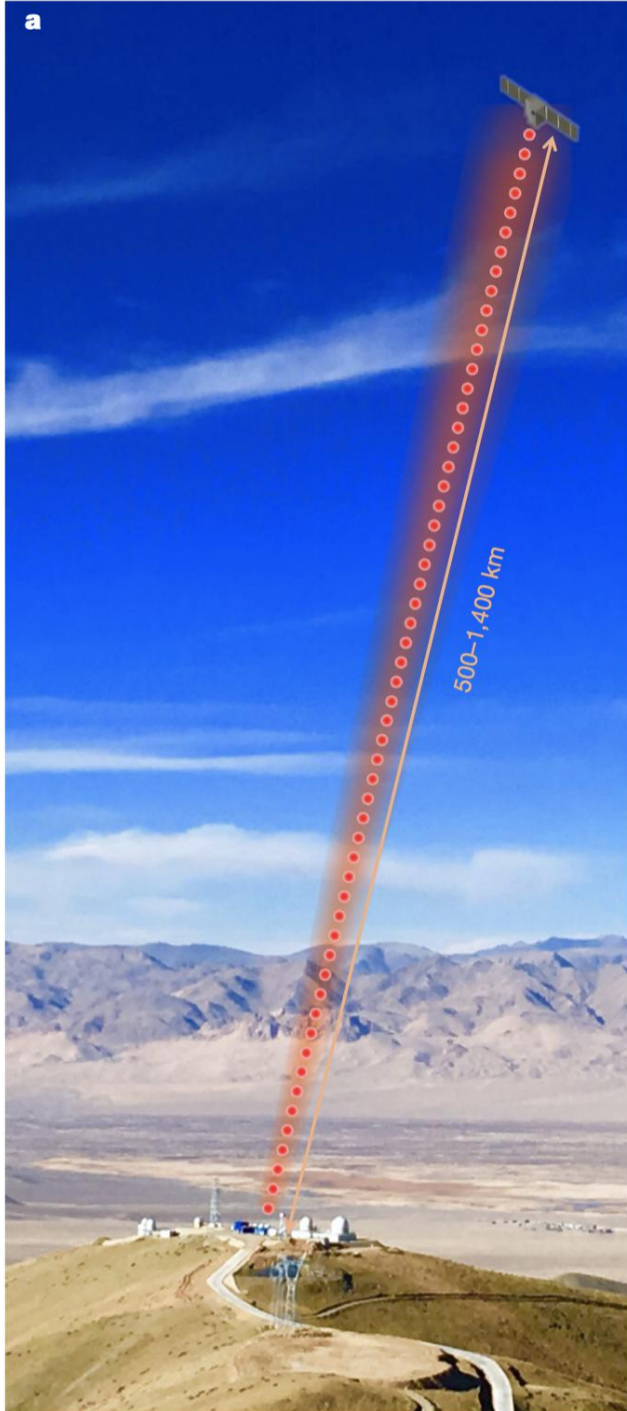
Ground-to-satellite quantum teleportation

Ji-Gang Ren^{1,2}, Ping Xu^{1,2}, Hai-Lin Yong^{1,2}, Liang Zhang^{2,3}, Sheng-Kai Liao^{1,2}, Juan Yin^{1,2}, Wei-Yue Liu^{1,2}, Wen-Qi Cai^{1,2}, Meng Yang^{1,2}, Li Li^{1,2}, Kui-Xing Yang^{1,2}, Xuan Han^{1,2}, Yong-Qiang Yao⁴, Ji Li⁵, Hai-Yan Wu⁵, Song Wan⁶, Lei Liu⁶, Ding-Quan Liu³, Yao-Wu Kuang³, Zhi-Ping He³, Peng Shang^{1,2}, Cheng Guo^{1,2}, Ru-Hua Zheng⁷, Kai Tian⁸, Zhen-Cai Zhu⁶, Nai-Le Liu^{1,2}, Chao-Yang Lu^{1,2}, Rong Shu^{2,3}, Yu-Ao Chen^{1,2}, Cheng-Zhi Peng^{1,2}, Jian-Yu Wang^{2,3} & Jian-Wei Pan^{1,2}

empty space. Here we report quantum teleportation of independent single-photon qubits from a ground observatory to a low-Earth-orbit satellite, through an uplink channel, over distances of up to 1,400 kilometres. To optimize the efficiency of the link and to counter the atmospheric turbulence in the uplink, we use a compact ultra-bright source of entangled photons, a narrow beam divergence and high-bandwidth and high-accuracy acquiring, pointing and tracking. We demonstrate successful quantum teleportation of six input states in mutually unbiased bases with an average fidelity of 0.80 ± 0.01 , well above the optimal state-estimation fidelity on a single copy of a qubit (the classical limit)¹⁴. Our demonstration of a ground-to-satellite uplink for reliable and ultra-long-distance quantum teleportation is an essential step towards a global-scale quantum internet.

首次实现地空之间的量子隐形传态。

向“全球尺度的量子互联网”迈进



Alice和纠缠源都在地面上

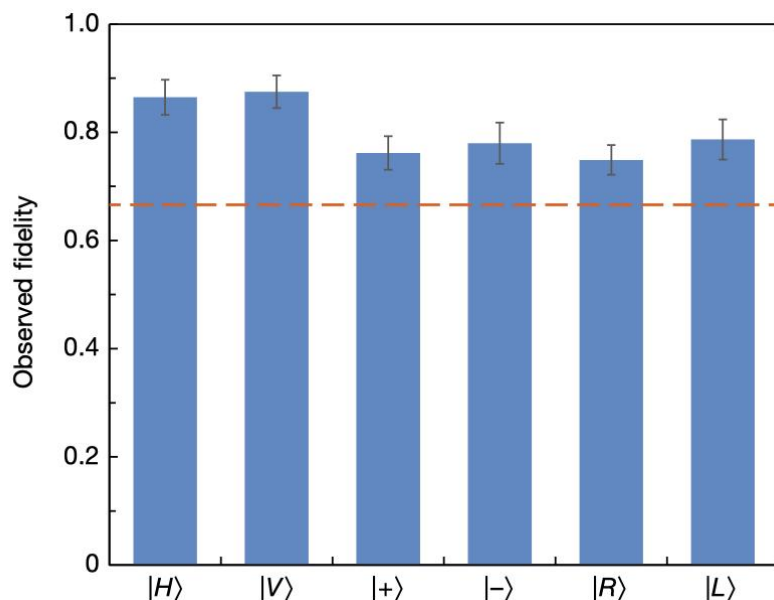
Bob在卫星上

thus projecting the wave function into a subspace of $|\phi^\pm\rangle_{12} = (|H\rangle_1|H\rangle_2 \pm |V\rangle_1|V\rangle_2)/\sqrt{2}$. Finally, by measuring the photons in the $(|H\rangle \pm |V\rangle)/\sqrt{2}$ bases at the outputs of the PBS, we can distinguish between $|\phi^+\rangle_{12}$ and $|\phi^-\rangle_{12}$ (ref. 17). After the Bell-state measurement, the final four-photon count rate measured on the ground

只能测四个bell中的两个

实验有50%概率测量到这两个Bell基矢，根据结果对第三个态进行操作（什么也不做，或者加一个pi相位）：

record the two datasets. After applying 0 or π phase shift, depending on the outcome state ($|\phi^+\rangle_{12}$ or $|\phi^-\rangle_{12}$) of the Bell-state measurement, during post-processed of the data, we calculate the teleportation fidelity



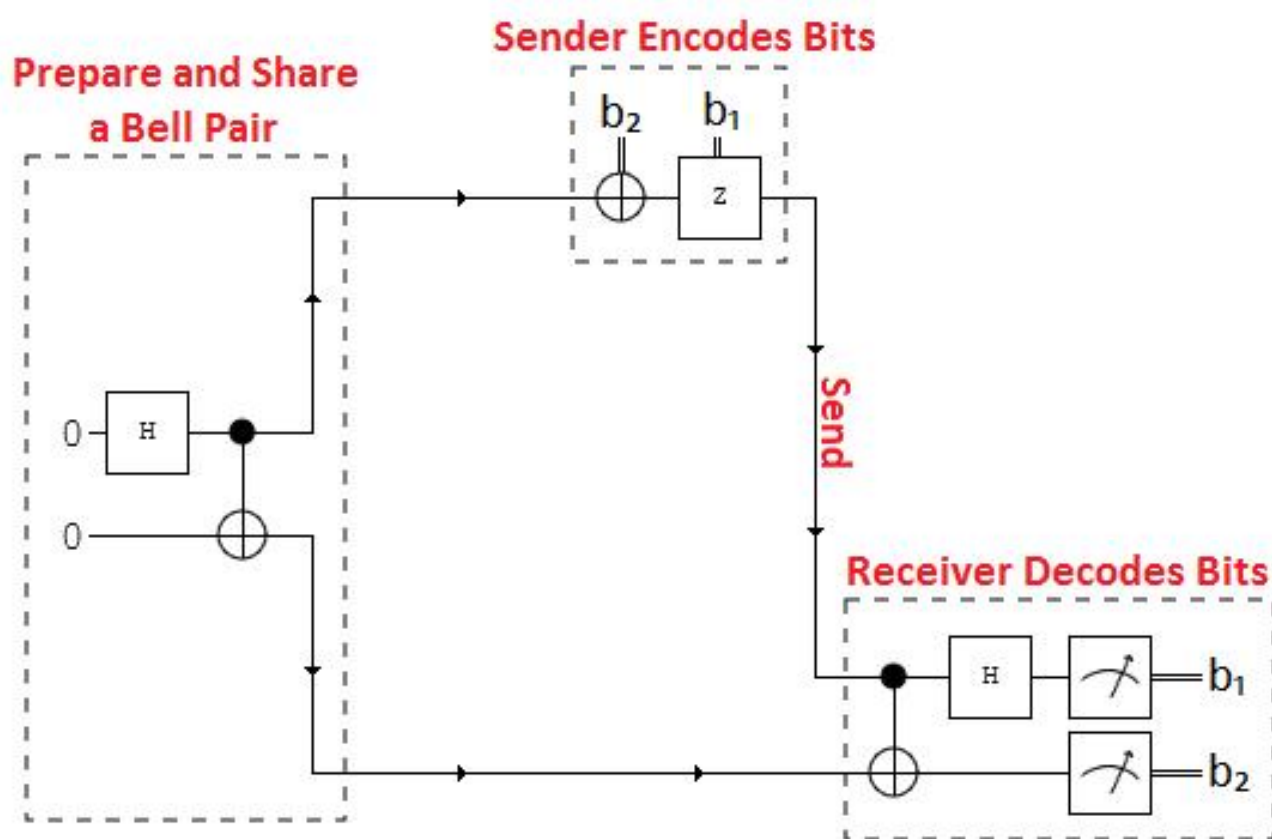
主要实验结果：

对于Bloch球上面的六个不同的态，测量第三个量子比特，给出与初态的保真度（交叠）。

1.10 示例：量子密集编码（量子超密编码）

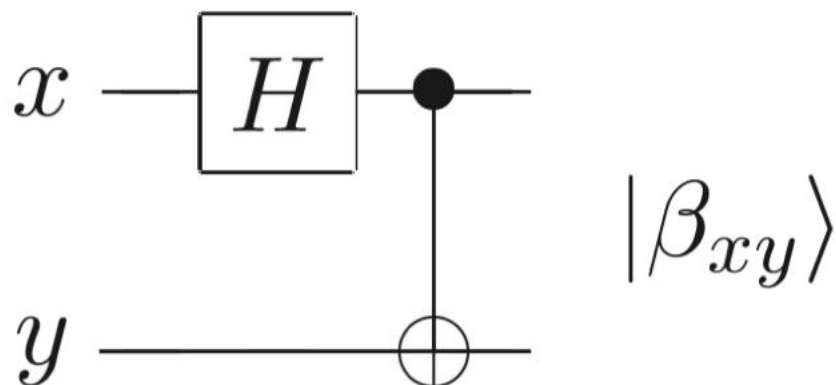
Alice想要传送“经典信息”给 Bob。

假设Alice拥有两个经典比特的信息想发送给Bob，但只允许向Bob发送一个量子比特。她能实现自己的目标吗？



回忆： 贝尔电路

由直积态出发，制备四个贝尔态。四个贝尔态互相正交，是可以区分的态。



$$|\beta_{xy}\rangle \equiv \frac{|0, y\rangle + (-1)^x |1, \bar{y}\rangle}{\sqrt{2}}$$

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}};$$

$$|\beta_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}};$$

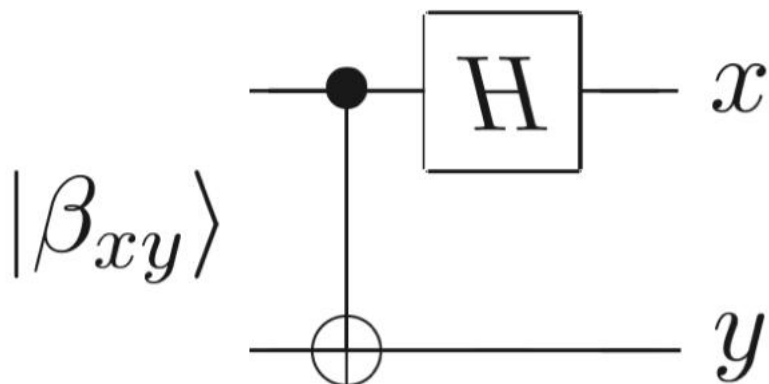
$$|\beta_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}};$$

$$|\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},$$

In	Out
$ 00\rangle$	$(00\rangle + 11\rangle)/\sqrt{2} \equiv \beta_{00}\rangle$
$ 01\rangle$	$(01\rangle + 10\rangle)/\sqrt{2} \equiv \beta_{01}\rangle$
$ 10\rangle$	$(00\rangle - 11\rangle)/\sqrt{2} \equiv \beta_{10}\rangle$
$ 11\rangle$	$(01\rangle - 10\rangle)/\sqrt{2} \equiv \beta_{11}\rangle$

贝尔电路的逆：反贝尔电路

由四个贝尔态出发，解码得到直积态



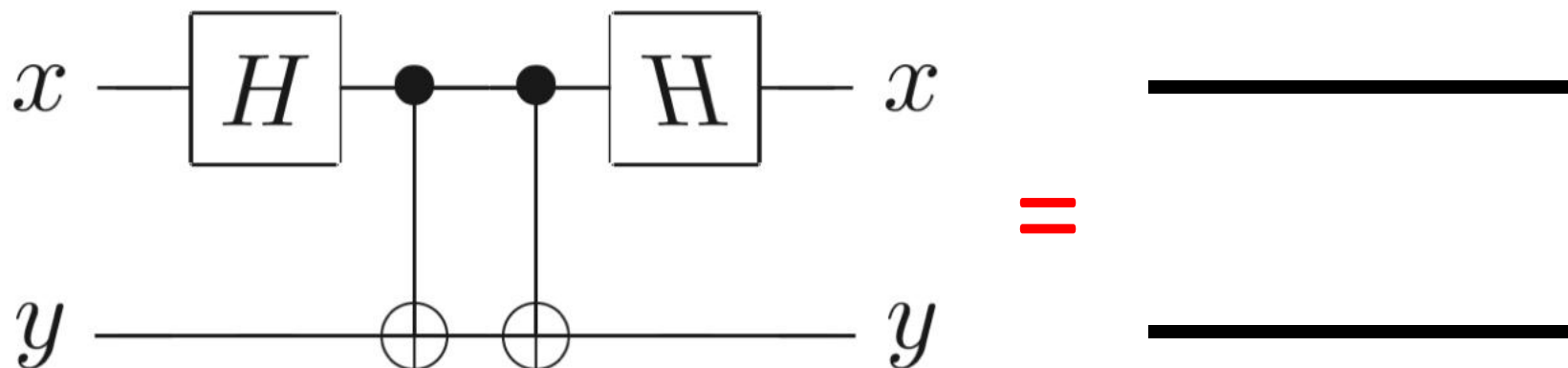
反贝尔电路的功能可以直接验证。（黑板上可以举例推导之。）

但如果我们不通过具体运算，如何得知这个“解码”过程的功能？

Out	In
In	Out
$ 00\rangle$	$(00\rangle + 11\rangle)/\sqrt{2} \equiv \beta_{00}\rangle$
$ 01\rangle$	$(01\rangle + 10\rangle)/\sqrt{2} \equiv \beta_{01}\rangle$
$ 10\rangle$	$(00\rangle - 11\rangle)/\sqrt{2} \equiv \beta_{10}\rangle$
$ 11\rangle$	$(01\rangle - 10\rangle)/\sqrt{2} \equiv \beta_{11}\rangle$

贝尔电路+反贝尔电路 = ?

什么都不做的电路

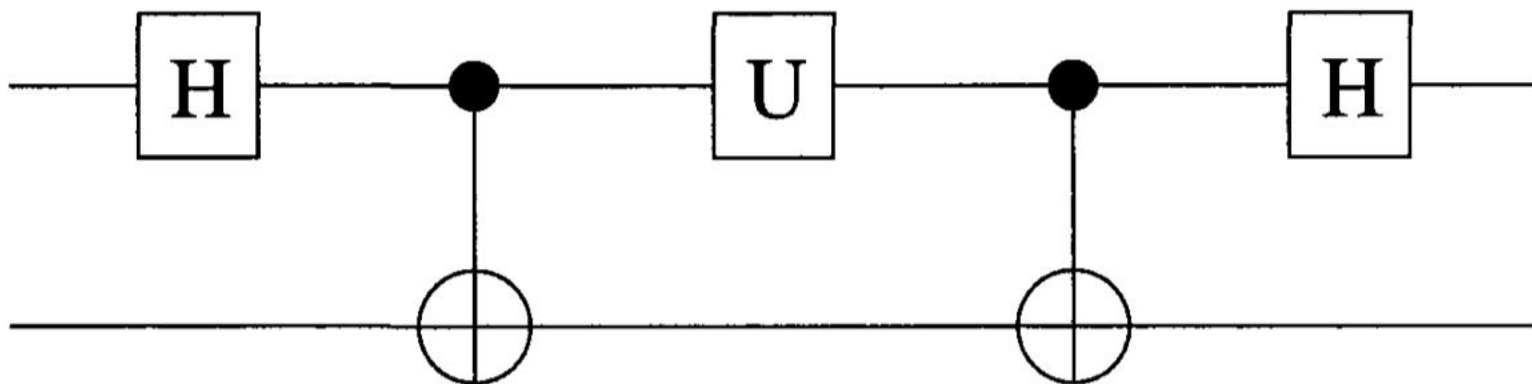


输入什么，就会输出什么。

这是因为 两个CNOT 门 的作用是什么都不做，
两个 H 门的作用也是什么都不做。

贝尔电路 + U + 反贝尔电路 = ?

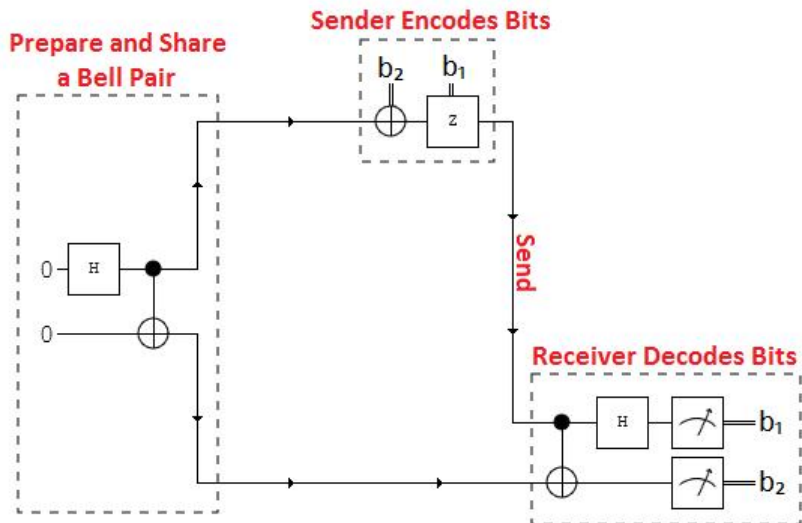
量子密集编码



问题：对于这个两比特的电路，作用在第一个比特上的独立操作有几种？

经典电路：两种 ($U = I$ 或 X ，即恒等门或者非门)

量子电路：四种 ($U = I, X, Z, XZ$)



1. 最初Alice与Bob共享纠缠对：

$$|\psi_1\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

2. 依据发送意向(b_2b_1)对Alice的量子比特做量子操作：

Alice	
00: I	01: Z
10: X	11: iY

2'. 量子操作的结果是生成了新的纠缠态（Bell基转换）：

$$|\psi_2\rangle = Z^{b_1} X^{b_2} |\psi_1\rangle$$

$$00 : |\psi\rangle \rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$01 : |\psi\rangle \rightarrow \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

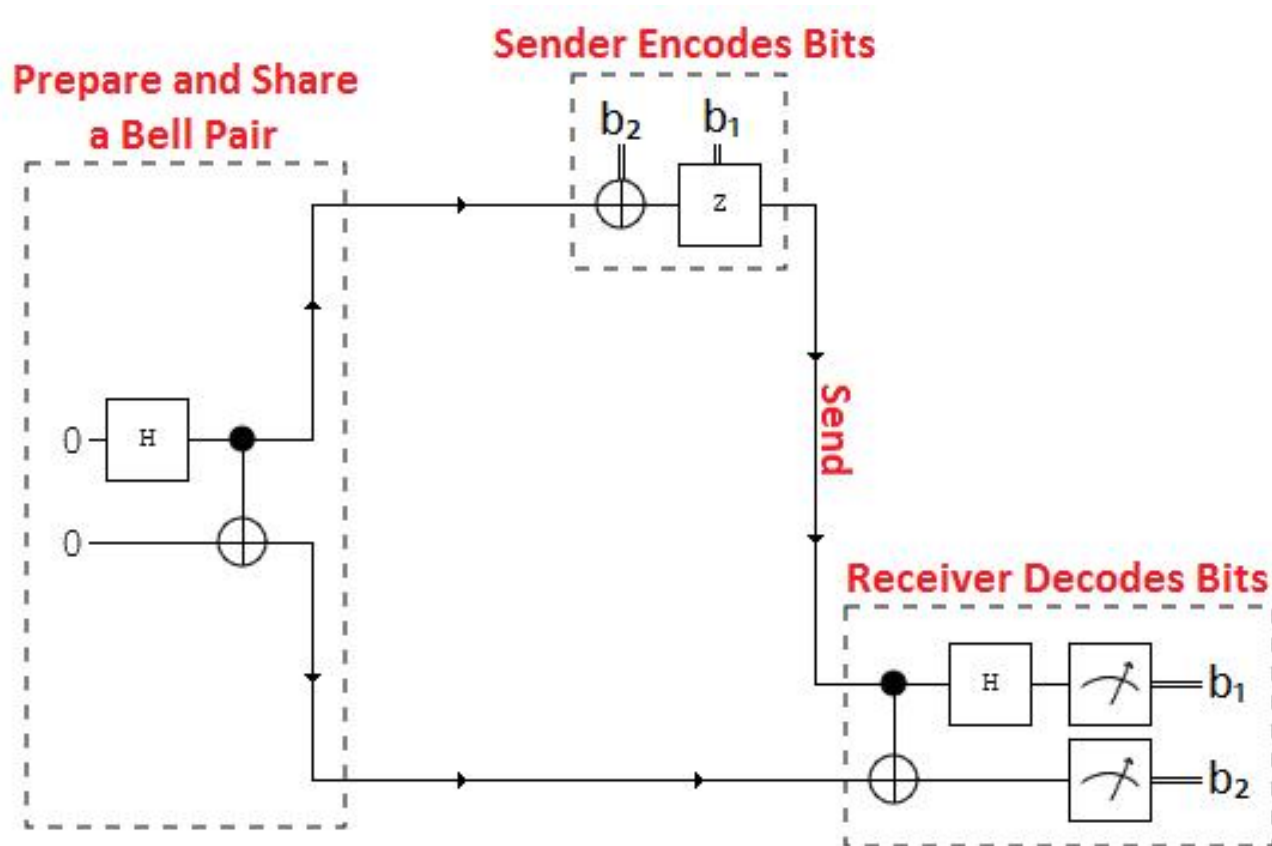
$$10 : |\psi\rangle \rightarrow \frac{|10\rangle + |01\rangle}{\sqrt{2}}$$

$$11 : |\psi\rangle \rightarrow \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

3. Bob 接受到Alice发送的量子比特，用反Bell电路解码得到(b_2b_1)。

量子密集编码的最终结果：

在Alice和Bob共享一个纠缠态的前提下（二人各持有贝尔态中的一个量子比特），Alice可以通过发送她的一位量子比特给Bob，由此向Bob传递两位经典信息。



1.10 示例：量子密集编码（量子超密编码）

实验解读：Zeilinger实验组首次实现了“部分功能” (1 trit)

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Dense Coding in Experimental Quantum Communication

Klaus Mattle,¹ Harald Weinfurter,¹ Paul G. Kwiat,^{1,2} and Anton Zeilinger¹

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(Received 22 November 1995)

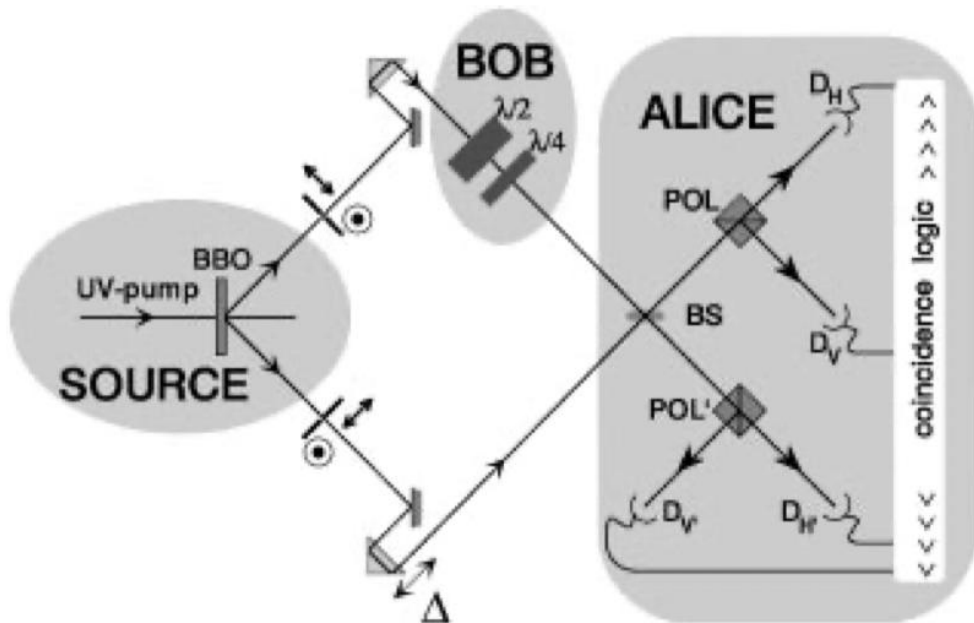
Classically, sending more than one bit of information requires manipulation of more than one two-state particle. We demonstrate experimentally that one can transmit one of three messages, i.e., 1 “trit” ≈ 1.58 bit, by manipulating only one of two entangled particles. The increased channel capacity is proven by transmitting ASCII characters in five trits instead of the usual 8 bits. [S0031-9007(96)00478-4]

two-particle interferometry can provide a solution to this problem [7]. Different interference effects allow one to identify two of the four Bell states, with the other two giving the same, third, measurement signal. Such an interferometric state analyzer therefore allows Alice to read three different messages sent via Bob's particle.

所以该系统可以区分三个态（共四个。）

1.10 示例：量子密集编码（量子超密编码）

实验解读：Zeilinger实验组首次实现了“部分功能”



1. 参量下转换生成贝尔态，一半发给Bob，另一半发给Alice。

2. Bob对自己的光子进行四种操作，随后发给Alice。

3. Alice对她的两个光子进行测量，看它们处于那个贝尔态，由此推断出Bob的操作是什么。

1.10 示例：量子密集编码（量子超密编码）

实验解读：Zeilinger实验组首次实现了“部分功能”

TABLE I. Overview of possible manipulations and detection events of the quantum dense coding experiment with correlated photons (we use h to denote the state of a photon in the mode towards detector D_H , etc.).

Bob's setting		State sent	State at output of Bell-state analyzer	Alice's registration events
$\lambda/2$	$\lambda/4$			
0°	0°	$ \Psi^+\rangle$	$\{hv + h'v' + vh + v'h'\}/2$	Coincidence between D_H and D_V or $D_{H'}$ and $D_{V'}$
0°	90°	$ \Psi^-\rangle$	$\{hv' - h'v' + v'h - v'h'\}/2$	Coincidence between D_H and $D_{V'}$ or $D_{H'}$ and D_V
45°	0°	$ \Phi^+\rangle$	$\{hh + vv + h'h' + v'v'\}/2$	2 photons in either D_H , D_V , $D_{H'}$, or $D_{V'}$
45°	90°	$ \Phi^-\rangle$	$\{hh - vv + h'h' + v'v'\}/2$	2 photons in either D_H , D_V , $D_{H'}$, or $D_{V'}$

Bob进行了四种操作
（对应于两个玻片
的四种不同偏振角
度组合。）

Alice可以判断对前面两种，
对第三、第四种操作得到
一样的结果。由此可推断
出Bob的操作是0、1、2。

1.10 示例：量子密集编码（量子超密编码）

实验解读：Zeilinger实验组首次实现了“部分功能”

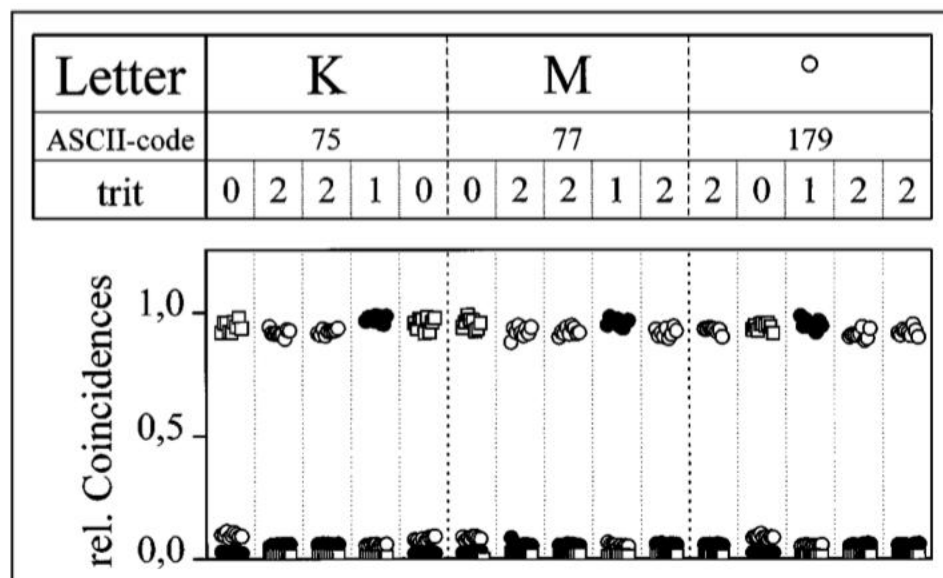


FIG. 5. “1.58 bit per photon” quantum dense coding: The ASCII codes for the letters “KM^o” (i.e., 75, 77, 179) are encoded in 15 trits (with “0” $\equiv |\Phi^- \rangle \hat{=} \square$, “1” $\equiv |\Psi^+ \rangle \hat{=} \bullet$, and “2” $\equiv |\Psi^- \rangle \hat{=} \circ$) instead of the 24 bits usually necessary. The data for each type of encoded state are normalized to the maximum coincidence rate for that state.

举例：Bob想向Alice发送一个字母K，其编码为0 2 2 1 0，他就将偏振玻片按序摆放即可。

Alice 通过测量结果可以推断出Bob玻片的摆放方式（经典信息）。



谢谢大家!



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